Unit III: Magnetic Effects of Current and Magnetism

CHAPTER - 4 MOVING CHARGE AND MAGNETISM

GIST OF THE CHAPTER:

Concept of magnetic field, Oersted's experiment,
Biot - Savart law and its application to the current carrying circular loop, Ampere's law and its applications to infinitely long straight wire, Straight solenoid (only qualitative treatment), Force on a moving charge in uniform magnetic and electric fields, Force on a current-carrying conductor in a uniform magnetic field, Force between two parallel current-carrying conductors-definition of ampere , Torque experienced by a current loop in uniform magnetic field, Current loop as a magnetic dipole and its magnetic dipole moment, Moving coil galvanometer - its current sensitivity and conversion to ammeter and voltmeter.

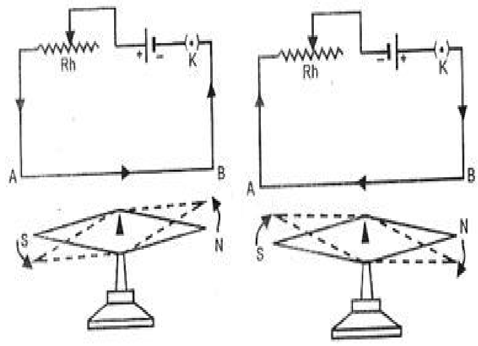
CONTENT/ CONCEPTS-

1. Magnetic field

A magnetic field is associated with an electric current flowing through a metallic wire. This is called the magnetic effect of current. On the other hand, a stationary electron produces an electric field only.

2. Oersted's experiment

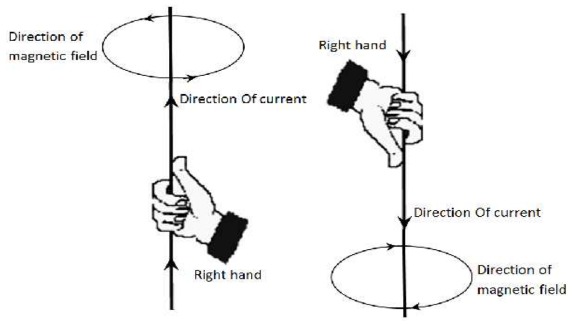
Hans christian Oersted performed simple experiment,
(i) When a needle is placed below the current carrying conductor it shows deflection.
(ii) When the direction of current is reversed the needle deflected in the opposite direction.
(iii) Deflection of the magnetic needle changes with the strength of electric current.



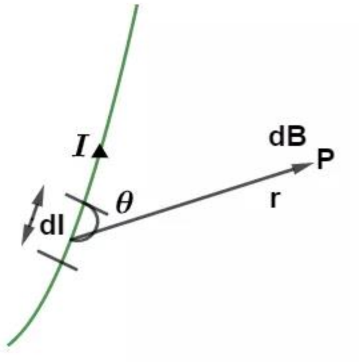
The SI unit of magnetic field is Wm^{-2} or T (tesla).
The strength of magnetic field is called one tesla, if a charge of one coulomb, when moving with a velocity of 1 ms^{-1} along a direction perpendicular to the direction of the magnetic field experiences a force of one newton.
1 tesla (T) = 1 weber $meter^{-2}$ (Wbm^{-2}) = 1 newton $ampere^{-1}$ $meter^{-1}$ (NA^{-1} m^{-1})
CGS units of magnetic field are called gauss or oersted.
1 gauss = 10^{-4} tesla.

3. Right hand thumb rule

Hold a conductor is Right Hand in such a way that thumb indicates the direction of current and curled finger encircling the conductor will give the direction of magnetic field lines.



4. Biot - Savart law



It states that the magnetic field strength $\overrightarrow{dB}$ produced due to a current element (of current I and length dl) at a point having position vector r relative to current element is-

$$\overrightarrow{dB} = \frac{\mu_0}{4\pi}\frac{I\,\overrightarrow{dl} \times \vec{r}}{r^3}$$

where μ_0 is the permeability of free space. Its value is -

$\mu_0 = 4\pi \times 10^{-7}$ Wb/A-m.

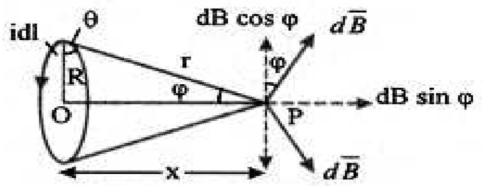
The magnitude of magnetic field is -

$$dB = \frac{\mu_0 I\ dl\ sin\theta}{4\pi r^2}$$

where θ is the angle between current element Idl and position vector $\vec{r}$ as shown in the figure. The direction of magnetic field $\overrightarrow{dB}$ is perpendicular to the plane containing Idl and $\vec{r}$.

5. Magnetic field due to a current carrying circular loop

The magnetic field due to current carrying circular coil of N-turns, radius a, carrying current I at a distance x from the center of coil is -



$$B = \frac{\mu_0 NIR^2}{2(R^2 + x^2)^{3/2}}$$

The magnetic field due to current carrying circular coil is along the axis.

At center, x=0

$$B_c = \frac{\mu_0 NI}{2R}$$

The direction of the magnetic field at the center is perpendicular to the plane of the coil.

6. Ampere's circuital law

It states that the line integral of magnetic field $\vec{B}$ along a closed BOUNDARY of an open surface is equal to μ_0-times the current (I) passing through the open surface with closed boundary.

$$\oint \vec{B}.\overrightarrow{dl} = \mu_0 I$$

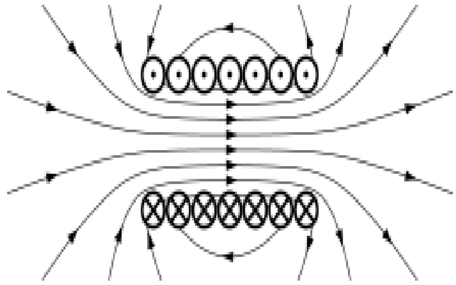
7. Magnetic field due to infinitely long straight wire using ampere's law

According to ampere's circulate law

$$\oint \vec{B}.\overrightarrow{dl} = \mu_0 I \Rightarrow \oint Bdl\,cos0^o = \mu_0 I$$

$$\Rightarrow \oint Bdl = \mu_0 I \ \Rightarrow B \times\ 2\pi r\ = \mu_0 I\ \Rightarrow B = \frac{\mu_0 I}{2\pi r}$$

8. Straight solenoid (only qualitative treatment)



At the axis of a long solenoid, carrying current I

$B = \mu_0 nI$, where n = N/L = number of turns per unit length

Tip - When we look at any end of the coil carrying current,

If the current is in anticlockwise direction then that end of the coil behaves like north pole, and

If the current is in clockwise direction then that end of the coil behaves like the south pole.

9. Force on a moving charge in uniform magnetic and electric fields

The force on a charged particle moving with velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$ is given by

$\overrightarrow{F_m} = q(\vec{v} \times \vec{B})$ = qvBsinθ

The direction of this force is determined by using Fleming's left hand rule.

The direction of this force is perpendicular to both $\vec{v}$ and $\vec{B}$,

When $\vec{v}$ is parallel to $\vec{B}$, then $\overrightarrow{F_m} = 0$

When $\vec{v}$ is perpendicular to $\vec{B}$, then $\overrightarrow{F_m}$ is maximum, i.e., $F_m = qvB$

The total force on a charged particle moving in simultaneous electric field $\vec{E}$ and magnetic field $\vec{B}$ is given by

$$\vec{F} = q(\vec{E} + \vec{V} \times \vec{B})$$

This is called the Lorentz force equation.

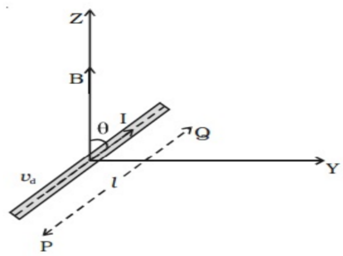
10. Force on a current-carrying conductor in a uniform magnetic field

$$\vec{F} = I(\vec{l} \times \vec{B})$$

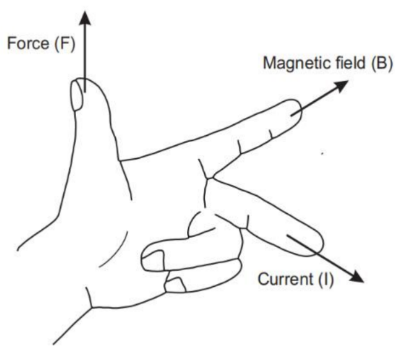
Magnitude of force is F = IlBsinθ

Direction of force $\vec{F}$ is normal to $\vec{l}$ and $\vec{B}$ given by Fleming's Left Hand Rule.

If $\theta = 0$ (i.e. $\vec{l}$ is parallel to $\vec{B}$), then the magnetic force is zero.



11. Fleming's Left Hand Rule



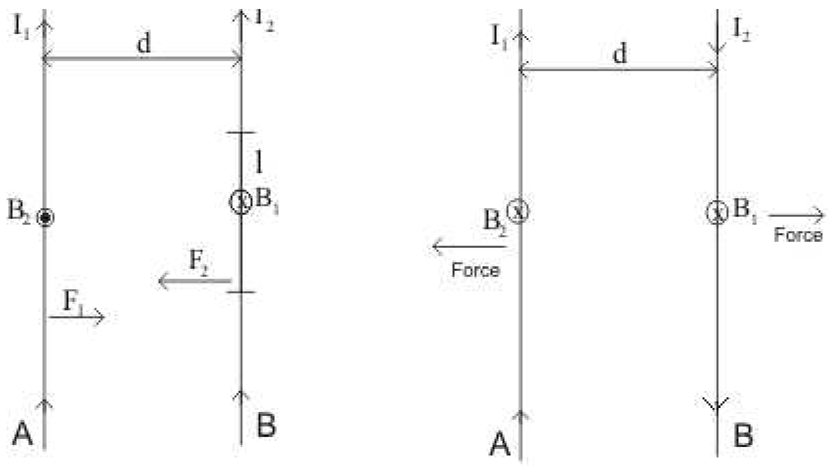
Stretch the left hand such that the fore-finger, the central finger and thumb are mutually perpendicular to each other. When the fore-finger points in the direction of the magnetic field and the central finger points in the direction of current then thumb gives the direction of the force acting on the conductor.

12. <u>Force between two parallel current-carrying conductors-definition of ampere</u>

Two parallel current carrying conductors attract while antiparallel current carrying conductors repel. The magnetic force per unit length on either current carrying conductor at separation 'a' is given by

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi a}$$

Its unit is newton/meter abbreviated as N/m



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Definition of ampere in SI system :- 1 ampere is the current which when flowing in each of the two parallel wires in vacuum at separation of 1 m from each other exert a force of $\frac{\mu_0}{2\pi} = 2 \times 10^{-7}$N/m on each other.

13. <u>Torque experienced by a current loop in uniform magnetic field</u>

$$\vec{\tau} = NI\left(\vec{A} \times \vec{B}\right) = \vec{M} \times \vec{B} = MB\ \sin\theta$$

Where $\vec{M} = NI\vec{A}$ is the magnetic moment of the loop.
The unit of magnetic moment in SI system is Am^2
The torque is maximum when the coil is parallel to the magnetic field and zero when the coil is perpendicular to the magnetic field.

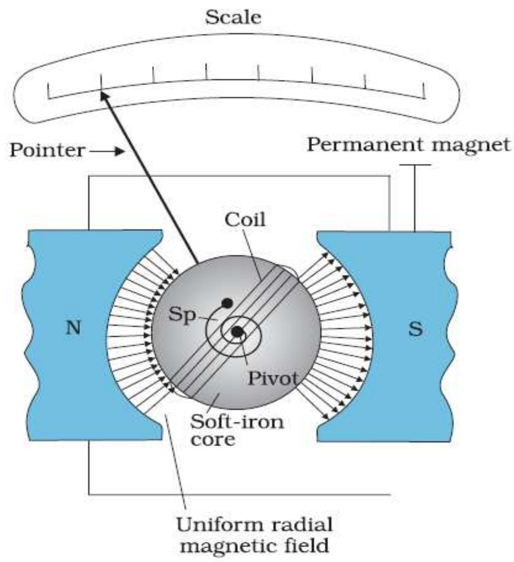
14. <u>Potential energy of a current loop in a magnetic field</u>

When a current loop of magnetic moment M is placed in a magnetic field, then potential energy of magnetic dipole is

$$U = -\vec{M}.\vec{B} = -MBcos\theta$$

(i) When θ=0, U = - MB (minimum or stable equilibrium position)
(ii) When θ=π, U = +MB (maximum or unstable equilibrium position)
(iii) When $\theta = \frac{\pi}{2}$, potential energy is zero

15. Moving coil galvanometer



A moving coil galvanometer is a device used to detect flow of current in a circuit. A moving coil galvanometer consists of a rectangular coil placed in a uniform radial magnetic field produced by cylindrical pole pieces. Torque on coil due to current-

τ = NIBA

where N is the number of turns, A is the area of coil. If k is torsional rigidity of material of suspension wire, then for deflection θ, torque-

$$\tau = k\theta$$

For equilibrium NIAB = kθ

$$\theta = \frac{NAB}{k} I \Rightarrow \theta \, \alpha \, I$$

Clearly, deflection in galvanometer is directly proportional to current, so the scale of galvanometer is linear.

Use of radial magnetic field :

The angle between the normal of the plane of loop and magnetic field $\theta = 90^o$, $\tau \propto I$, when radial magnetic field is used the deflection of coil is proportional to the current flowing through it. Hence a linear scale used to determine the deflection of coil.

Uses of galvanometer :

(i) Used to detect electric current is a circuit.
(ii) Used to convert the ammeter by putting a low resistor.
(iii) Used to convert voltmeter by putting a high resistor.
(iv) Used as ohmmeter by making special arrangement

Figure of Merit of a galvanometer: - The current which produces a deflection of one scale division in the galvanometer is called its figure of Merit. It is equal to $k = \frac{E}{(R+G)\theta}$

Sensitivity of a galvanometer:-

(i) **Current sensitivity:** It is defined as the deflection of coil per unit current flowing in it.

Current Sensitivity $I_S = \frac{\theta}{I} = \frac{NAB}{k}$

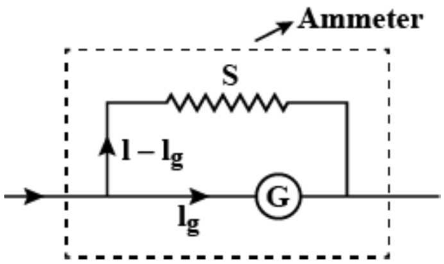
(ii) **Voltage sensitivity:** It is defined as the deflection of coil per unit potential difference across its ends

Voltage sensitivity $V_S = \frac{\theta}{IR} = \frac{NAB}{kR}$

16. Conversion of Galvanometer into Ammeter :-

A galvanometer may be converted into ammeter by using very small resistance in parallel with the galvanometer coil. The small resistance connected in parallel is called a shunt.

If G is resistance of galvanometer, I_g is current in galvanometer for full scale deflection, then for conversion of galvanometer into ammeter of range I ampere, the shunt is given by

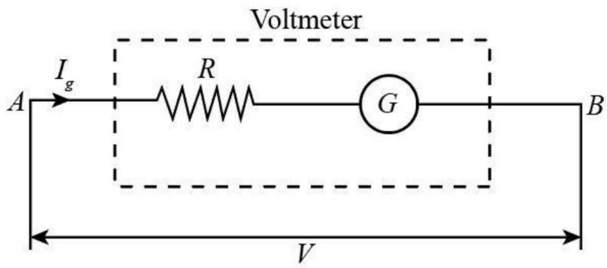


$$S = \frac{I_g}{I - I_g} G$$

Resistance of ideal ammeter is zero.

17. Conversion of Galvanometer into Voltmeter :-

A galvanometer may be converted into voltmeter by connecting high resistance (R) in series with the coil of the galvanometer. If V volt is the range of voltmeter formed, then series resistance is given by



$$R = \frac{V}{I_g} - G$$

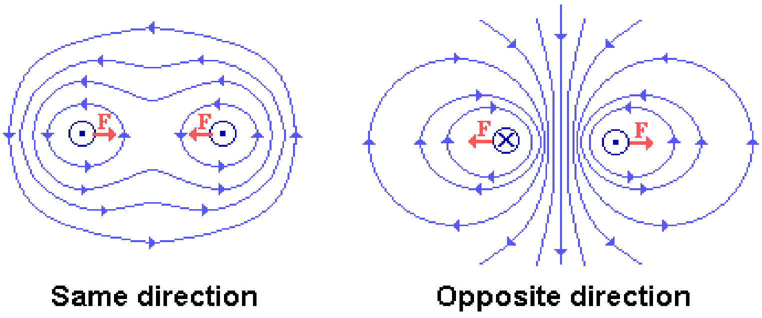
Resistance of ideal voltmeter is infinite.

UNITS :-

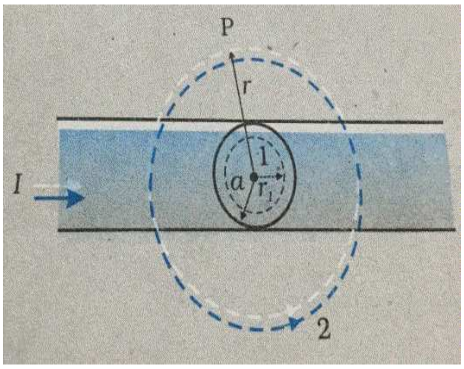
Physical quantity	Symbol	Nature	Dimensions	Units	Remarks
Permeability of free space	μ_0	scalar	$[MLT^{-2}A^{-2}]$	Henry/meter Or TmA^{-1}	$4\pi \times 10^{-7}$
Magnetic field	B	vector	$[MT^{-2}A^{-1}]$	T (tesla) or N/A-m or Wb/m^2	
Magnetic moment	M	vector	$[L^2A]$	$A\text{-}m^2$ or J/T	
Torsion constant	K	scalar	$[ML^2T^{-2}]$	$N\text{-}m\text{-}rad^{-1}$	
Torque	τ	vector	$[ML^2T^{-2}]$	N-m	

GRAPHS :-

1. Field pattern for force between two parallel current carrying conductors -



2. Magnetic field in the region r < a and r > a, for a long straight wire of a circular cross-section



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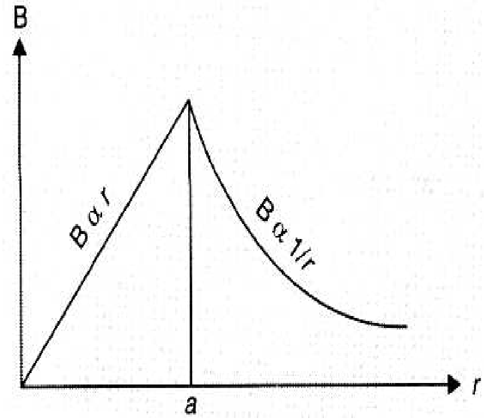
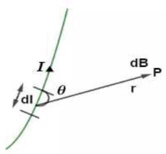
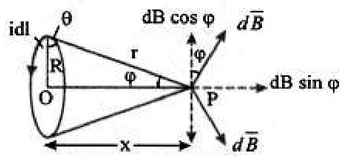
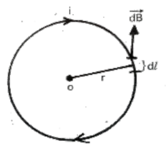
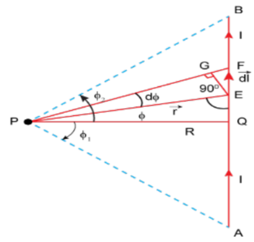
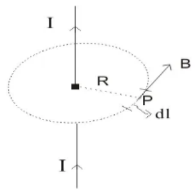
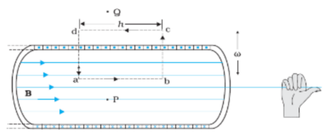
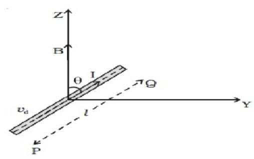
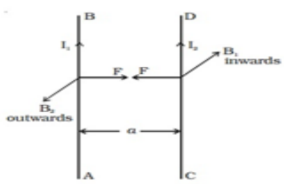
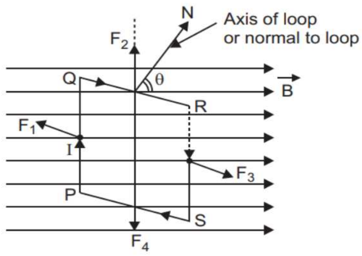
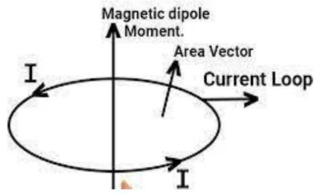
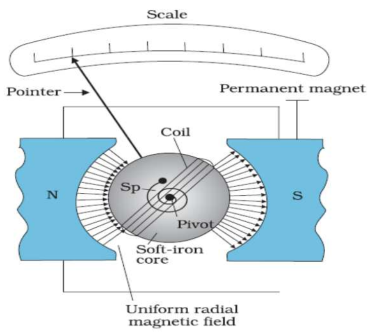
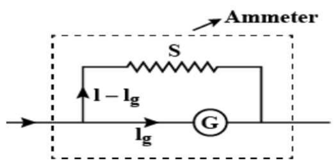


A long straight wire of a circular cross-section

FORMULAE & DIAGRAMS :-

Sr. No.	Term	Formula	Diagram
1	Biot Savart's Law	$dB = \dfrac{\mu_0 I\ dl\ sin\theta}{4\pi r^2}$	
2	Magnetic field at a point on the axis of a current carrying circular coil	$B = \dfrac{\mu_0 NIR^2}{2(R^2 + x^2)^{3/2}}$	

3	Magnetic field at a center of a current carrying circular coil	$B = \frac{\mu_0 I}{2r}$	
4	Magnetic field due to a current in a straight conductor	$B_0 = \frac{\mu_0}{4\pi}\frac{I}{r}(sin\phi_1 + sin\phi_2)$	
5	Ampere's circuital law	$\oint \vec{B}.\vec{dl} = \mu_0 I$	
6	Magnetic field due to an infinitely long current carrying straight wire	$B = \frac{\mu_0 I}{2\pi R}$	
7	Magnetic field due to straight solenoid	B = μ_0 n I , where n = N/L n = number of turns per unit length	
8	**Lorentz force** Force on a moving charge in uniform magnetic and electric fields	$\vec{F} = q(\vec{E} + \vec{V} \times \vec{B})$	
9	Force on a current-carrying conductor in a uniform magnetic field	$\vec{F} = I(\vec{l} \times \vec{B})$ $F = BIl\, sin\theta$	
10	Force between two parallel current-carrying conductors	$F = \frac{\mu_0 I_1 I_2 l}{2\pi a}$	

11	Torque experienced by a current loop in uniform magnetic field	$\tau = NIBA\,sin\theta$ $\vec{\tau} = \vec{m} \times \vec{B}$ Where m = IA = magnetic moment	
12	Current loop as a magnetic dipole and its magnetic dipole moment	$B = \frac{\mu_0}{4\pi}\frac{2m}{x^3}$ Where x is the distance along the axis from the center of the loop m = NIA = NIπr^2	
13	Moving coil galvanometer -	1. $I = G\theta$ Where $G = \frac{k}{NAB} =$ galvanometer constant K = torsional N = number of turns A = area of coil B = magnetic field θ = deflection	
14	Moving coil galvanometer Current sensitivity Voltage sensitivity	$I_s = \frac{\theta}{I} = \frac{NAB}{k}$ $V_s = \frac{\theta}{IR} = \frac{NAB}{kR}$	
15	Conversion of galvanometer to ammeter	$S = \frac{I_g}{I - I_g}G$ G = galvanometer resistance	
16	Conversion of galvanometer to voltmeter	$R = \frac{V}{I_g} - G$ G = galvanometer resistance R = high resistance in series